

[I225] Statistical Signal Processing(E) Office Hour 4

1. Consider a continuous-time LTI smoothing filter defined by

$$Y(t) = \frac{1}{2T} \int_{t-T}^{t+T} X(\tau) d\tau.$$

Suppose the input random process $X(t)$ has power spectral density

$$S_{XX}(\omega) = \frac{1}{1 + \omega^2}.$$

- (a) Find the impulse response $h(t)$.
 - (b) Find the frequency response $H(\omega)$.
 - (c) Find the output power spectral density $S_{YY}(\omega)$.
 - (d) Explain whether the filter behaves as a low-pass or high-pass filter.
2. Consider the following first-order differential equation system with white noise input:

$$Y'(t) + 2Y(t) = W(t).$$

The input white noise has autocorrelation

$$R_{WW}(\tau) = 4\delta(\tau).$$

- (a) Find the transfer function $H(\omega)$.
 - (b) Find the output power spectral density $S_{YY}(\omega)$.
 - (c) Find the autocorrelation function $R_{YY}(\tau)$.
 - (d) Determine whether the system is causal or non-causal.
3. You are measuring a constant DC voltage θ using a noisy voltmeter. Every time you take a measurement, you get the true value θ plus some Gaussian noise w . You take N independent measurements:

$$x[n] = \theta + w[n], \quad n = 0, 1, \dots, N-1$$

The noise $w[n]$ is zero-mean white Gaussian noise where the samples are I.I.D (independent and identically distributed) with a known variance σ^2 .

Supposed you decide to estimate the voltage by simply taking the average of all measurement:

$$\hat{\theta} = \frac{1}{N} \sum_{n=0}^{N-1} x[n]$$

(e) Show that this estimator is unbiased.

(f) Compute the variance of $\hat{\theta}$ and check whether $\hat{\theta}$ is consistent.

4. A sensor measures the distance to a target $x[n]$. Due to a specific type of quantization error, the measurements are uniformly distributed between 0 and an unknown maximum distance θ :

$$p(x; \theta) = \begin{cases} \frac{1}{\theta}, & 0 \leq x \leq \theta \\ 0, & \text{otherwise} \end{cases}$$

Find the Cramér-Rao Lower Bound (CRLB) of θ , if it is a valid boundary.

5. A sensor counts the number of packets $x[n]$ arriving at a network switch every second. The counts are independent and follow a Poisson distribution with an unknown arrival rate λ :

$$p(x[n]; \lambda) = \frac{e^{-\lambda} \lambda^{x[n]}}{x[n]!}$$

After some time, N independent counts have been observed.

Find the Cramér-Rao Lower Bound (CRLB) of λ , if it is a valid boundary.